

Linux & Operating Systems — In-Depth Question Paper

Processes, Context Switching, Shell Scripting, Scheduling & Linux Fundamentals

Instructions:

1. This paper contains 25 multiple-choice questions divided into four sections.
2. Each question carries 2 marks. Total marks: 50.
3. Choose the single best answer (a, b, c, or d) for each question.
4. No negative marking. Attempt all questions.

Section A — Process Management & Context Switching

(2 marks each)

1. How is a “process” most accurately described within the context of an operating system?
 - a) A file saved on the disk drive.
 - b) A sequence of CPU instructions.
 - c) A running instance of a program, along with its current state and resources.
 - d) A reserved section of memory for an active program.
2. Under what circumstance does a process move from the “Running” state into the “Waiting” state?
 - a) When it finishes executing entirely.
 - b) When a process with greater priority preempts it.
 - c) When it needs to pause for an I/O operation to finish.
 - d) When it is assigned CPU time to execute.
3. Which data structure does the operating system rely on to hold all the information about a specific process?
 - a) The Stack
 - b) The Process Control Block (PCB)
 - c) The Program Counter
 - d) The Heap
4. What role does the Process Control Block (PCB) primarily serve?
 - a) Storing the compiled machine code of a program.
 - b) Handling memory allocation and release.
 - c) Acting as a container for all the data required to manage one particular process.
 - d) Facilitating communication between processes.
5. When a context switch takes place, which step happens first?
 - a) The incoming process's saved state is restored from its PCB.
 - b) The outgoing process's state (registers, program counter, etc.) is written to its PCB.
 - c) The scheduler picks which process runs next.
 - d) The CPU's cache memory is cleared.
6. What does the overhead associated with context switching mainly consist of?
 - a) Time spent waiting on an I/O request.
 - b) Time used to save and restore process states.
 - c) Time the CPU spends running the user's application code.
 - d) Time devoted to handling interrupts.
7. If a process is currently in the “Ready” state, what does that indicate?
 - a) It is actively running on the CPU right now.
 - b) Its execution has already concluded.
 - c) It is on hold, waiting for an I/O event.
 - d) It resides in memory, waiting for its turn on the CPU.

8. Which item is normally found inside a Process Control Block (PCB)?
- a) The disk's physical address.
 - b) A full copy of the process's memory contents.
 - c) The process's current state along with its CPU register values.
 - d) The program's original source code.
9. How does a system process, such as a scheduler, differ from an ordinary user process?
- a) System processes are limited to running only in the "Waiting" state.
 - b) System processes run in a more privileged mode and can access restricted hardware.
 - c) User processes are always assigned higher priority.
 - d) User processes don't have an associated PCB.
10. What capability does context switching primarily provide?
- a) Spawning a new process from an existing one.
 - b) Blocking a process from reaching unauthorized memory.
 - c) Making a single CPU appear to run several processes at once.
 - d) Ending a process when an error occurs.

Section B — Linux & Shell Fundamentals

(2 marks each)

11. What is a defining difference that sets Linux apart from Unix?
- a) Linux only offers a command line, whereas Unix includes a GUI.
 - b) Linux is open-source and can be freely distributed, unlike proprietary Unix.
 - c) Multi-user support exists in Unix but not in Linux.
 - d) Linux predates Unix in its development.
12. Which command lists the files and folders inside a directory on Unix/Linux?
- a) cd
 - b) mkdir
 - c) ls
 - d) pwd
13. In a Unix-like operating system, what is the shell mainly responsible for?
- a) Managing hardware peripherals directly.
 - b) Providing a graphical desktop environment.
 - c) Interpreting the commands a user types and running programs accordingly.
 - d) Storing the user's saved files.
14. Which statement correctly compares "for" loops to "while" loops in shell scripting?
- a) "for" loops handle conditional branching, and "while" loops iterate over lists.
 - b) "for" loops repeat a set number of times or over a list, while "while" loops keep going as long as a condition holds true.
 - c) "while" loops always execute faster than "for" loops.
 - d) "for" loops work only with numeric ranges.
15. Which command switches the current directory to the logged-in user's home directory?
- a) cd /
 - b) cd ..
 - c) cd ~
 - d) cd home
16. Which statement about shell scripts is accurate?
- a) They need to be compiled prior to running.
 - b) They consist of a series of commands saved in a file, ready to be executed.
 - c) Bash is the only shell language they can be written in.
 - d) They're mainly meant for building graphical applications.

17. Which keyword marks the start of a conditional statement in shell scripting?

- a) then
- b) fi
- c) if
- d) else

Section C — CPU Scheduling & Synchronization

(2 marks each)

18. What is CPU scheduling primarily trying to achieve?

- a) Reducing the total number of active processes.
- b) Getting the most out of the CPU while ensuring processes get fair access.
- c) Stopping all processes from executing at the same time.
- d) Distributing memory among processes.

19. Which of the following is a well-known CPU scheduling algorithm?

- a) Depth-First Search (DFS)
- b) First-Come, First-Served (FCFS)
- c) Binary Search
- d) Bubble Sort

20. Why is synchronization needed in operating systems?

- a) So multiple processes can share resources without causing inconsistent data.
- b) To make a single process run faster.
- c) To convert data between formats.
- d) To handle network connectivity.

Section D — System Administration & Commands

(2 marks each)

21. While installing CentOS 8 Stream, which step is essential for the system to work correctly afterward?

- a) Staying offline throughout the setup.
- b) Setting up a root password and creating a user account.
- c) Installing every available package by default.
- d) Skipping disk partitioning entirely.

22. Which command would create a new folder called my_folder?

- a) touch my_folder
- b) rmdir my_folder
- c) mkdir my_folder
- d) cp my_folder

23. What does running the man command generally accomplish in Unix/Linux?

- a) Creates a brand-new file.
- b) Shows the manual/documentation pages for a given command.
- c) Restarts the machine.
- d) Deletes a directory.

24. Which loop type keeps executing as long as its condition evaluates to false?

- a) for loop
- b) while loop
- c) until loop
- d) foreach loop

25. What was one of the main motivations behind creating Linux?

- a) To build a closed, proprietary operating system.

- b) To offer a free, open-source alternative to commercial Unix systems.
- c) To run exclusively on mainframes.
- d) To be the very first OS with a graphical interface.

Answer Key

Q#	Answer	Q#	Answer
1	C	2	C
3	B	4	C
5	B	6	B
7	D	8	C
9	B	10	C
11	B	12	C
13	C	14	B
15	C	16	B
17	C	18	B
19	B	20	A
21	B	22	C
23	B	24	C
25	B		